NHL Honours Project Archival Audit

Source verified findings and reproduction status
Companion to the 2026 thesis revision

Howard R. Silvey
Research edition 0.1.0
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Original honours project supervised by Dr. Jason Nielsen
Carleton University School of Mathematics and Statistics
Original submission April 7, 2020

Independent 2026 research revision and extension. This edition is separate from the original university submission and does not imply institutional review or supervisor endorsement of the new analysis.

Primary sharing format PDF
Editable companion DOCX
Repository github.com/howardsilvey7/nhl-goal-prediction

Audit conclusion

The 2020 honours project survives as a complete 47-page PDF. It is sufficient to verify the broad research design and principal reported numbers. Exact execution cannot yet be reproduced because the local recovery inventory has not found the original NHL data table or executable R analysis.

The major methodological problem is the mismatch between an individual-game forecasting objective and a team-season estimation sample. The revision preserves the original record and corrects the unit, target, information boundary and validation design in a separately labeled modern experiment.

This report distinguishes what the source says, what has been independently checked by inspection, what remains ambiguous and what must be recovered before an exact archival reproduction can be claimed.

Contents

1 Archival identity and preservation

2 Source recovery and evidence status

3 Numerical reconciliation

4 Methodological corrections

5 Data dictionary corrections

6 Reproduction plan

7 Relation to the new experiment

References

The PDF includes bookmarks for the chapter headings. Table and figure captions identify the numbered research outputs.

1 Archival identity and preservation

Author in the original title page: Howard Silvey. Supervisor: Dr. Jason Nielsen. Institution: Carleton University, School of Mathematics and Statistics. Submission date: April 7, 2020. Title: National Hockey League Data: Analysis and Prediction.

The 2026 author identity is Howard R. Silvey. The new work is titled National Hockey League Goal Prediction, with the subtitle Modern Statistical and Machine Learning Approaches and a revision-and-extension statement. The 2020 submission date is not applied to the new analysis.

The local archival PDF is copied without rewriting metadata, re-pagination or repairs. Its public Carleton URL is retained as provenance. The accidental suffix in the original public filename is not treated as a scientific version identifier.

Preserved artifact: archive/Silvey-2020-NHL-Honours-Project.pdf. Pages: 47. SHA256: a3102150485b5d81aae0af29f844124cd7e8607f2349271d2cb3c0271260b6d0.

2 Source recovery and evidence status

The recovery scope for this edition is a targeted local filename inventory of University materials, Documents and Desktop Downloads. Matching NHL, hockey, honours, R source/history, spreadsheet and archive names were considered. No original NHL executable source or complete input table was recovered. The surviving PDF is an original artifact; the supplied 2026 baseline is a later derivative record.

The Documents R history contains recent archive-extraction commands for a different project and does not supply NHL analysis. Unrelated private files were not copied into the research repository. This inventory is not a claim that every cloud drive, email attachment, removable device or older computer was searched.

The PDF evidence permits source verification, not rerunning the analysis. Reproduction should remain unresolved until surviving original materials are found or their absence is explicitly established. A reconstruction is an acceptable scientific fallback only when its assumptions and discrepancies are disclosed.

Artifact or finding	Classification	Current status
Original 2020 PDF	Original	Preserved and checksummed
September 2026 baseline	Derivative audit	Preserved with source attribution
Original R scripts	Unknown	Not found in targeted local search
Original team-season table	Unknown	Not found in targeted local search
Historical headline metrics	Reported original results	Visually source-verified, not recomputed
Modern game dataset and models	New 2026 analysis	Separate empirical experiment

3 Numerical reconciliation

Physical PDF page numbers below count the cover as page 1. The document's printed chapter pagination begins later and therefore differs. The main LOOCV screenshot is on physical page 39, printed page 32.

The table dimensions support 123 team-season rows and approximately 40 candidate columns. The source's 4,920 figure counts table cells, not the number of independent observations. The cross-validation display itself states 123 samples, five predictors and training samples of size 122.

The backward model retains CF, CA, CF_Percent, PP_Percent and L. Its reported R squared is 0.1161. The forward model's reported R squared is 0.06451. The source discusses adjusted R squared but the final console values and selection path still need exact code-level reconciliation; this audit does not infer a full coefficient table from narrative alone.

Claim	Verified source	Execution status
123 estimation samples	Physical pages 17 and 39	Original rows unavailable
Backward R squared 0.1161	Physical pages 28 and 32	Not recomputed
Forward R squared 0.06451	Physical page 32	Not recomputed
RMSE 46.20022	Physical page 39 screenshot	Not recomputed
R squared 0.03927991	Physical page 39 screenshot	Not recomputed
MAE 37.91543	Physical page 39 screenshot	Not recomputed

The LOOCV values concern the original season-total response. Dividing an RMSE by 82 would yield a rescaled season-average error, not a validated individual-game forecasting error. Modern regulation-goal metrics must not be presented as a direct before-and-after reduction on the same task.

4 Methodological corrections

Unit and target. Replace team-season totals with one paired record per game. Define the two goal counts explicitly and separate regulation, overtime and shootout conventions. The paired-response idea is already anticipated by the original abstract.

Leakage. Exclude target-game and same-period outcome-derived features. Lag shooting, special-teams, save, standings and scoring measures. A valid feature can be unsafe when its time window overlaps the target period.

Validation. Replace pooled random sampling and LOOCV with ordered expanding windows and tuning inside the training period. Team repetition and changing scoring environments require attention beyond a simple independence assumption.

Outcome support. OLS can produce negative point estimates and supplies no count distribution without additional assumptions. Retain an historical OLS benchmark where reconstruction becomes possible; use count models for coherent probability forecasts.

Selection and collinearity. CF, CA and CF percentage are algebraically related. Other totals and ratios may introduce rank deficiency or unstable estimates. NA coefficients require a rank and preprocessing audit. Stepwise p values are not protected from repeated-search uncertainty.

Inference and uncertainty. State OLS assumptions precisely; do not use the Central Limit Theorem to justify all residual properties. Separate coefficient inference, point forecasting and distribution forecasting. Report uncertainty for model differences rather than a universal acceptable RMSE threshold.

Schedule exposure. A fixed 82-game divisor is inappropriate for incomplete or disrupted seasons. Rates must use actual exposure, or the response should be defined directly at game level.

Claims. Two correctly rounded forecasts in seven displayed cases are illustrative, not a stable general accuracy estimate. The revision evaluates every admitted held-out game and retains negative results.

5 Data dictionary corrections

FA is labeled inconsistently in the original glossary and appears intended to represent Fenwick Against. The high-danger percentage label and formula need reconciliation. Save percentage must use saves rather than an ambiguous S already defined as team shots.

PDO should distinguish its shooting and save components and specify its scale. Penalty-kill percentage requires an explicit opponent power-play scoring numerator and denominator. Totals, rates, percentages and adjusted statistics must be marked separately.

The revised registry should include canonical name, formula, unit, side, aggregation level, source, window, lag rule, earliest availability, missing-value policy and transformation lineage. These definitions make a predictor auditable rather than merely recognizable to a hockey reader.

Original issue	Required archival check	Revision treatment
FA label	Inspect source column and glossary	Canonical for/against naming
High-danger percentage	Recover exact inputs/formula	No unverified formula carried forward
SV percentage ambiguity	Identify actual saves numerator	Explicit saves/shot exposure definition
PDO description	Verify scale and component sums	Separate definitions and availability
Home goals versus team total	Recover response construction	Paired regulation target
Incomplete schedule	Recover extraction cutoff and games played	Game counts and exclusions explicit

6 Reproduction plan

First inventory surviving source files without modification. Calculate hashes, record original names and provenance, identify restricted materials and map every artifact to thesis tables and figures. Recovery can include a user-supplied archive without renaming its internal contents.

Restore the R interpreter and packages as closely as evidence permits. Preserve the original analysis separately from any corrected re-analysis. If exact environment restoration is impossible, record version substitutions and assess their numerical impact.

Verify the 123 rows, season labels, extraction cutoff and approximately 40 columns. Recover the true response, exposures, import types, missing values and deterministic feature identities. Reproduce the full OLS fit and inspect aliased coefficients before trying to match selection output.

Recover both selection procedures, significance thresholds, exact formulas and coefficients. Reconstruct the three displayed prediction samples and determine whether the data used for each were genuinely excluded from estimation.

Re-run caret LOOCV with selection and preprocessing placement documented. Reconcile each headline metric and every material discrepancy. A discrepancy ledger must state whether it comes from source transcription, a different sample, altered preprocessing, selection behavior or environment change.

If original inputs cannot be recovered, build a clearly labeled reconstruction from equivalent historical sources. Never relabel reconstruction as reproduction. The absence of original files limits archival claims but does not invalidate a separately designed modern study.

7 Relation to the new experiment

The modern study uses 10,969 paired regular-season records, reconstructed regulation goals and conservative earlier-date features. It implements nine variants with five development seasons and a separately frozen final season. Its results belong to the 2026 extension.

The initial empirical result is a modest proper-score improvement from regularized recent-form predictors. The shared bivariate component is zero and the point-error gain is small. These results neither reproduce nor numerically correct the original LOOCV output.

The archival status remains source-verified but not execution-reproduced. The project status ledger and release notes retain that qualification. This report is complete as an audit of available evidence, while the reproduction task remains open.

References

[1] Silvey, H. (2020). National Hockey League Data: Analysis and Prediction. Undergraduate honours project, Carleton University, School of Mathematics and Statistics. Supervisor: Dr. Jason Nielsen. April 7, 2020. https://carleton.ca/math/wp-content/uploads/Howard-Silvey-Honours-Project-2.pdf

[11] Silvey, H. R. (2026). NHL Honours Project 2020 Audit and 2026 Research Baseline. Version 2.0, September 27, 2026. Preserved transfer document in docs/TRANSFER_BASELINE.md.

[12] Silvey, H. R. (2026). NHL Goal Prediction research repository. Research edition 0.1.0. https://github.com/howardsilvey7/nhl-goal-prediction